

MODULE 1: DIGITAL DEVICES

Introduction to Digital Devices

Digital devices are pieces of hardware that use computers or microcontrollers, and they are found everywhere in our digital world. They enhance and support how we live our lives every day. They can connect and work together to give us the data we need, when and where we need it. Digital devices are always developing, changing the way in which they are used by individuals, organisations, and local, national, and global societies.

Types of Digital Devices

- **Computers**
- **Mobile phones**
- **Tablets**
- **Cameras and camcorders**
- **Home entertainment systems**
- **Home automation devices and smart assistants**
- **Navigation aids**

What is a Computer?

A computer is an electronic machine that inputs data, processes it under the control of a stored program, and outputs information. While the data is being processed, it can be retrieved from or saved on backing storage.

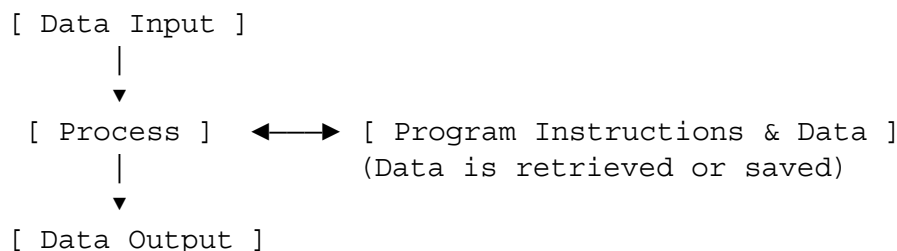
- **Data:** Raw, unprocessed information.
- **Information:** Data that people understand. In order to understand data, you must interpret it.

Example of Data vs. Information: The number **30112012** is raw data. It can be interpreted as:

- A date: 30/11/2012
- A sum of money: \$301,120.12

The interpreted date or sum of money represents the **information**.

The Computer Information Processing Model



- **Input:** Data is entered into the computer.

- **Process (Turn data into information):** The computer is controlled by a program (a sequence of instructions). It processes the input data automatically following these instructions.
- **Store:** The program and data are stored (e.g., on a hard disk) for later retrieval.
- **Output:** The computer communicates to the user (e.g., displaying graphics on a screen).

Types of Computers

- **PCs (Personal Computers):** A microcomputer designed for individual use.
- **Mainframe Computers:** Large computers with massive processing power, designed to do a lot of complicated jobs quickly for many users at the same time.
- **Minicomputers:** A smaller version of a mainframe computer.
- **Supercomputers:** A very large, ultra-powerful mainframe.

Different Types of Personal Computers (PCs)

- **Desktop PC:** Consists of basic components including a monitor, keyboard, system unit, and mouse.
 - *All-in-One PC:* A variation that combines the monitor together with the internal computer hardware into a single unit.
- **Laptop PC:** A portable computer slightly larger than A4 paper size.
 - *Desktop Replacements:* Larger laptops featuring higher screen resolutions and performance capabilities than standard portable computers.

Embedded Computers

An embedded computer is designed for and built into a specific application where it will perform a limited range of dedicated functions. The size and functionality depend entirely on the application. They can be tiny devices built into a single microchip to control hardware like digital watches, satellites, or media systems.

Hand-Held Computers

A handheld computer, Personal Digital Assistant (PDA), or palmtop can fit in one hand or a pocket. It is generally too small for general text-heavy work. A PDA usually has a touch-sensitive screen and uses a stylus. While they can be temporarily attached to a physical keyboard, they are not designed for comfortably typing long documents.

Mobile Devices & Connectivity

Mobile Phones

- Mobile phones utilize a **SIM (Subscriber Identity Module)** card to connect to mobile network infrastructures.
- Specialist phones offer customized features to meet specific, niche user requirements.

Smartphones

- Small computers with integrated Wi-Fi and mobile network connectivity for calling and internet access.
- Combine the features of multiple separate devices: cameras, media players, and handheld games consoles.
- Feature advanced operating systems capable of downloading **Applications (apps)** to customize features for entertainment, education, and business.
- Rely on a touch screen and **virtual keyboard** for text input.
- *Drawback:* Combining so many features uses significant power, resulting in shorter battery lives requiring regular recharging.

Tablets

- Larger than smartphones but utilize similar features (touch screens, apps, and Wi-Fi connectivity).
- Some versions feature built-in SIM card slots to maintain mobile data connections when outside the range of a Wi-Fi signal.

Cameras and Camcorders

- Digital cameras and camcorders use light sensors to capture images formed by light passing through the device's lens.
- Traditionally, cameras capture still images and camcorders capture moving images.
- Modern convergence allows most digital cameras to film video and most camcorders to shoot high-quality still photos.

Home Entertainment & Smart Systems

Home Entertainment Components

- **Televisions:** Display still and moving images. Image quality is determined by the number of small color dots (**pixels**) making up the screen. This is known as the screen's **resolution**.
- **Sound Systems:** Produce loud, rich audio using high-quality speakers and amplifiers (some speakers include built-in amplifiers).
- **Media Players:** Electronic devices that store and play digital music, videos, and photos.
- **Personal Video Recorders (PVR):** Devices that record broadcasted content so it can be viewed at a later date.
- **Blu-ray and DVD Players:** Connect to televisions to read and play films stored on physical disks.
- **Games Consoles:** Designed to let users play video games on a television screen using wireless controllers. Advanced systems utilize motion sensors, virtual reality (VR) controllers, and headsets to track real-world movements and replicate them via an avatar.

Home Automation and Smart Assistants

An emerging technology group used to build **smart homes**. Home automation systems connect a range of digital devices that sense and control household functions like temperature and

lighting. These systems can be controlled remotely via smartphone apps or over the internet.

Navigation Aids

Navigation devices (such as Sat-Navs) calculate optimal routes between locations and dynamically update directions if a turn is missed. They are commonly used in cars, delivery vans, and ships.

- They use information from **GPS (Global Positioning System)** satellites to track precise coordinates on Earth.
- Integrated **orientation sensors** detect which direction the device faces to align the on-screen map.
- **Note on Connectivity:** GPS functions directly via satellite radio signals without needing the internet. However, navigation aids require an active internet connection to download live map data and real-time traffic updates.

Key Features of Digital Devices

Feature	Description
Portability	How easily a device can be carried or moved around; directly determined by its physical size and weight.
Performance	The speed at which a device completes tasks. Dictated by processor instruction speed and RAM (Random Access Memory) capacity. More RAM allows more programs to run simultaneously.
Storage	The capacity where files and programs are saved. Higher storage speeds also improve overall system performance.
Connectivity	Sharing data via wired or wireless options. Wired connections are faster and more reliable but carry extra costs and physical tripping hazards.
Media Support	The ability to read/write data to external media like SD, microSD cards, or DVDs. External adapters can add support if native slots are missing.
Expansion	Capability to add components via internal slots or external ports (e.g., USB ports for connecting external peripherals).
Energy	Electricity requirements. Lower consumption in mobile devices extends battery life, cuts costs, and reduces environmental impact.

Security Features

Software Security

Designed to prevent unauthorized access to private, valuable user data. Common software security access checks include:

- Passwords
- PINs (Personal Identification Numbers)
- Biometric systems

Physical Security

Crucial to prevent physical device theft. Devices incorporate hardware security slots that allow physical steel cable locks to secure the machine firmly to office furniture.

User Interfaces (UI)

The user interface controls how a human interacts with and issues commands to a digital device.

- **Command Line Interface (CLI):** The user types explicit text instructions into a prompt, and the system responds strictly with text feedback. It uses minimal memory but requires learning specific commands.
- **Menu-Driven Interface:** Displays commands as a structured list or menu options (used in ATMs, TVs, and older mobile phones). Sweeping through multiple menus takes more time than typing a direct command, but requires no command memorization.
- **Graphical User Interface (GUI):** A visual system controlled by an on-screen pointer utilizing **WIMP** structures: **W**indows, **I**cons, **M**enus, and **P**ointers. It is the easiest interface to navigate but consumes the largest amount of memory and storage.
- **Voice Interface:** Allows spoken voice commands. Uses voice recognition software to match spoken words against an internal or cloud-based library. Requires internet access if the vocabulary library is hosted online. Background noise or accents can sometimes cause incorrect matches.
- **Gesture Interface:** Allows users to control a screen by swiping, pinching, or moving fingers directly across a touch-sensitive panel.

MODULE 2: PERIPHERAL DEVICES

What is a Peripheral Device?

A peripheral device is any hardware component that can be connected externally or internally to a computer host system (like a PC or tablet). They are classified into three distinct functional categories:

1. **Input Devices**
2. **Output Devices**
3. **Storage Devices**

Input Peripherals

Input peripherals send data from the outside environment into the computer system, allowing users to control software or capture external data.

1. Keyboards

Keyboards use buttons called keys that users press to enter alphanumeric characters or execute shortcuts. They communicate data to the computer through either wired cables or wireless frequencies.

2. Pointing Devices

Used to manipulate an on-screen pointer to select items:

- **Mouse:** Employs an optical sensor to track movement across a flat surface. Legacy variants use a mechanical rolling trackball.
- **Tracker Ball:** A stationary base containing an exposed ball that the user rolls directly with their fingers or palm. It does not require moving the whole device body.
- **Track Pad:** A flat, touch-sensitive surface that senses finger friction, tapping, and multi-touch gestures. Commonly built into laptops to save physical space.

3. Gaming & Art Inputs

- **Joysticks:** Levers commonly integrated into gaming controllers to steer on-screen characters or graphic objects called **sprites**.
- **Graphics Tablet:** A flat, pressure-sensitive pad used in conjunction with an electronic pen called a **stylus**. Highly favored by digital artists, illustrators, and architects.

4. Scanners

Scanners employ light sensors to convert physical paper documents into flat digital image files.

- **Optical Character Recognition (OCR):** Specialized software that scans a document's image, identifies printed text characters, and translates them into an editable text document.
- **Optical Mark Recognition (OMR):** Software used alongside scanners to capture simple checkbox marks on a predefined form. It is universally used to grade multiple-choice examinations quickly.

5. Barcode Scanners

Reads patterns of alternating black lines and white spaces by detecting their respective widths. These patterns correspond to numbers or letters that identify a specific item. Used extensively for retail sales and parcel tracking.

- **Linear Barcodes:** Standard one-dimensional striped codes.
- **Matrix Barcodes (QR Codes):** Modern two-dimensional codes that can hold vastly more information than linear codes and can be scanned from any rotational angle.

6. Webcams & Microphones

- **Webcams:** Specialized digital cameras that capture video feed for internet streaming or security monitoring. They are generally lower resolution than dedicated digital camcorders.
- **Microphones:** Devices that capture audio waves using a flexible physical diaphragm.

Sound waves cause the diaphragm to vibrate, producing a continuous analog electrical signal. The signal passes through the computer's sound card, which functions as an **Analog-to-Digital Converter (ADC)** to create an editable digital audio file.

Touch Screen Technology

Touch screens serve as both input and output elements. There are two principal methodologies:

- **Resistive Touch Screens:** Constructed of two flexible layers separated by a microscopic gap. When a finger presses down, the layers touch, closing an electrical circuit.
 - *Pros:* Very durable and highly resistant to dust/water.
 - *Cons:* Harder to read under bright lights due to multiple reflective layers, and cannot register multi-touch gestures (only one press at a time).
- **Capacitive Touch Screens:** Features a layer of electrical charge-storing material underneath a protective glass top layer. Because humans are naturally conductive, touching the glass draws away a tiny amount of electrical charge to the finger. The system calculates the exact coordinates of the drop in charge.
 - *Pros:* Supports multi-touch inputs and remains highly legible. Commonly utilized in modern smartphones.

Biometric Scanners

Biometric hardware evaluates unique human biological traits. To setup, a user is scanned multiple times to establish a statistical mathematical average, which is saved securely. Future entry attempts are verified against this record.

- **Fingerprint Recognition:** Analyzes individual patterns of loops, whorls, and arches on fingertips.
- **Facial Recognition:** Identifies structural geometric features of the human face.
- **Voice Recognition:** Uses a microphone to parse spoken phrases, matching the vocal waveform print against an authorized master copy.
- **Iris Recognition:** Scans the complex patterns within the iris of the eye. An iris scan is roughly **120 times more detailed** than a fingerprint, making it 120 times more secure and reliable.

Pros & Cons of Biometrics

- **Advantages:** Users do not need to memorize passwords (biometric features cannot be forgotten), transactions become much faster, scanners bypass language barriers, and they are extremely difficult for criminals to trick.
- **Disadvantages:** Raises severe privacy concerns regarding electronic storage of identity data, systems are highly expensive to implement, and some users feel physically uncomfortable being scanned.

Card Readers

Card readers intercept data written to portable plastic cards to authorize payments, grant building access, or track logistics.

- **Magnetic Strip:** Stores static data on a magnetic band. This is the least secure method because criminals can easily copy ("skim") the data without the owner's knowledge.

- **Programmable Chip (Chip & PIN):** Data stored within an integrated circuit on the card is encrypted and can only be unlocked when the user inputs the corresponding PIN into the terminal.
- **RFID & NFC:** Short-range wireless technologies. **Near Field Communication (NFC)** is a specialized extension of **Radio Frequency Identification (RFID)**. Small, cheap RFID tags containing a unique database ID number can be embedded in cards, key fobs, or phones for contactless transactions.

Environmental Sensors

Sensors automatically register real-world physical and chemical properties without needing human interaction, outputting data to a computer system for processing.

Advantages of Sensors

- Can be deployed in hazardous, remote, or unreachable environments.
- Operates continuously 24/7 without fatigue.
- Eliminates risks of manual human recording errors.
- Can detect phenomena outside human biological capabilities (e.g., micro-changes in pressure, toxic gas presence).

Target Factors Monitored

- Light
- Moisture and humidity
- Temperature
- Motion and movement
- Pressure



HOMEWORK ASSIGNMENT

(Answer the following questions based on the revision material covered above)

1. State one benefit of using barcodes in a supermarket.
2. State one reason why small, built-in microphones are used in smartphones and laptops.
3. Explain why capacitive touch screens do not work when you are wearing woolen gloves.
4. State three computer systems that do not use monitors.
5. State two benefits of using laser printers rather than inkjet printers.
6. State two types of business that use plotters.
7. Explain why some 3D printers use biodegradable materials to create models.
8. List two sensors used in smartphones. For each one, describe one way in which it is used by the operating system or an app.
9. Discuss the advantages and disadvantages of using sensors in smart environments.
10. Explain one reason why Blu-ray disks are used to store modern console games.